

QlikView

Mobile Clients

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1 Summary of QlikView clients

A QlikView client is required for display and usage of an existing QlikView document (.qvw file). While the QlikView Server is responsible for opening, hosting, and calculating the document, the clients are required for user interaction and presenting the document and its objects.

The choice of which client or set of clients to use is entirely dependent on the customer's environment and preferences. Client choices range from a fully installed QlikView Developer to a (no installation required) AJAX Zero-Footprint Client (ZFC), to a Mobile Client on your iPhone. Any combination of client types is allowed, as long as the proper licensing CALs are available (see Section QVS Licensing) on the server license. Finally, certain considerations must be taken into account when deploying a QlikView document with QlikView Server and its clients. See Section Considerations when developing documents for use with QlikView Server for information on limitations when using a QlikView client to interface with a QlikView document.

The following table provides a brief summary description of client choices.

QlikView Client	Description
Windows Installed Client	Full image desktop installed QlikView Desktop. Installation and client licensing required.
QlikView IE Plug-in	Full image ActiveX plug in for Internet Explorer web browser. Installation required. No client side licensing required.
QlikX Analyzer for Internet Explorer	Object only ActiveX plug in for Internet Explorer web browser. Requires web page design or QlikView Client Generator. Installation required. No client side licensing required.
AJAX Zero-Footprint Client (ZFC)	Object only Dynamic HTML client utilizing AJAX architecture in web browser. Web page is automatically generated by QlikView Server. No installation or client side licensing required.
QlikView iPhone client	One object-at-a-time view of full QlikView document. Download and install App through App Store. No client side licensing required.
QlikView BlackBerry client	Full image client is based on Java Mobile Edition (Java ME) and will run on most mobile phones that support MIDP 2.0. Download and install App on device. No client side licensing required.
QlikView Java Mobile clients	Full image client is based on Java Mobile Edition (Java ME) and will run on most mobile phones that support MIDP 2.0. Download and install App on device. No client side licensing required.

2 Requirements for Mobile Clients

A web server is needed to serve the mobile downloads for QlikView. In addition to the server requirements above, the mobile clients must be running compatible software. Client requirements are as follows:

Apple iPhone / iPod Touch 2G and 3G:

- Firmware 2.0 or above
- Wi-Fi or 3G recommended

BlackBerry

- OS 4.5 or later
- Wi-Fi or 3G recommended

Java Mobile Devices

- CLDC 1.1, MDIP 2.0 and JSR-172
- Wi-Fi or 3G recommended

3 The QlikView iPhone Client

General

The QlikView iPhone Client is available for free and can be downloaded from the Apple iTunes AppStore. Download and install the QlikView for iPhone app. The QlikView iPhone Client requires version 8.5 SR2 or later of QlikView Server. However, the client cannot be used with the Small Business edition of QlikView Server since the client uses the same communication protocol (QVPX) as the AJAX client.

Supported Devices

All generations of iPhone and iPod Touch are supported, using firmware version 3.0 or later.

Setup

Once the QlikView for iPhone app is downloaded and installed on your device, start the app, select the Settings menu and select **Edit** on the first screen. Here, you can set the URL, provide a Username and Password, and choose whether to use the Mobile Directory of the QlikView Server (generally recommended for documents specific to Mobile devices). If the Password is left out here, you will be required to enter it each time you open a document. The URL can be specified in four different ways (the qvp protocol is not supported):

- demo.qlikview.com
- http://demo.qlikview.com
- https://demo.qlikview.com
- the ip address

If you want your data/user credentials sent encrypted, you will need to set up a VPN for secure access to your server environment.

Document Access

If you wish to use anonymous access for you QlikView iPhone clients you must configure your server according to one of the following:

Anonymous Access Using NTFS Authorization

- Set the QlikView Server authorization to **NTFS** in QEMC on the tab **System, Setup, Security**.
- Set the QlikView Server authentication to **Always Anonymous** or **Allow Anonymous** in QEMC on the tab **System, Setup, Security**.
- Set the QlikView Web Server Ajax access to **Always Anonymous** in QEMC on the tab **System, Setyp, Ajax**.
- Give the user IQVS_MachineName access rights to the document on the file server.

Anonymous Access Using DMS Authorization

- Set the QlikView Server authorization to **DMS** in QEMC on the tab **System, Setup, Security**.
- Set the QlikView Server authentication to **Always Anonymous** or **Allow Anonymous** in QEMC on the tab **System, Setup, Security**.

- Set the QlikView Web Server Ajax access to **Always Anonymous** in QEMC on the tab **System, Setyp, Ajax**.
- Add the anonymous user to the document in QEMC in the **Edit User** dialog on the tab **Documents, User Documents, Authorization**.

If you wish to use authenticated access for your QlikView iPhone clients you must configure your server according to the following. Note that the client will not send any user credentials unless prompted for them.

Authenticated Access Using NTFS Authorization

- Set the QlikView Server authorization to **NTFS** in QEMC on the tab **System, Setup, Security**.
- Set the QlikView Server authentication to **Allow Anonymous** or **Prohibit Anonymous** in QEMC on the tab **System, Setup, Security**.
- Unmark the check box **Always Anonymous** for the QlikView Web Server Ajax access in QEMC on the tab **System, Setyp, Ajax**.
- Give the user in question access rights to the document on the file server.

Authenticated Access Using DMS Authorization

- Set the QlikView Server authorization to **DMS** in QEMC on the tab **System, Setup, Security**.
- Set the QlikView Server authentication to **Allow Anonymous** or **Prohibit Anonymous** in QEMC on the tab **System, Setup, Security**.
- Unmark the check box **Always Anonymous** for the QlikView Web Server Ajax access in QEMC on the tab **System, Setyp, Ajax**.
- Add the user in question to the document in QEMC in the **Edit User** dialog on the tab **Documents, User Documents, Authorization**.

Application Flow

The list of documents is created based on the contents of the root folder and the mounted folders on the QlikView Server. Setting the **Use Mobile Directory** option limits the list to the documents in the subfolder **Mobile** of the QlikView Server root folder.

Once you have selected a QlikView document to open, you will be presented with a navigation bar with the Sheets available in the document. Selecting a sheet will then display one object on that sheet at any time. Other objects are displayed through cover flow at the bottom of the screen. The logic behind the order in which objects are displayed in the cover flow is: list boxes, charts, tables, text objects. The order within each group is arbitrary. All objects are, of course, updated as selection state changes. Standard iPhone touch interaction is supported for the object.

- Touch a value to select, touch it again to deselect. To select multiple values in a list box, swipe sideways on the list box. Two buttons are displayed at the top of the list box, **Cancel** and **Select**. Tap the values in the list box you wish to alter selection state for. The values will be marked with a tick. Tap again to remove the tick-mark. Tap **Select** to send the request to the Server or **Cancel** to go back to previous select state. Tapping an excluded (gray) value will include it in the current selection, if it is not in conflict with selections made in other list boxes. In case of a conflict, all conflicting values will be deselected before selecting the tapped value.
- For a listbox, the All/Possible/Excluded buttons can be used to make multiple selections.
- To make selections in charts, touch and hold for one second in the area you wish to select. A square with movable corners and two buttons is displayed. Tap and drag the corners to mark the area you wish to select. Tap **Select** to select the marked area or **Cancel** to keep the current selection.
- Use the **E-mail** button in a chart to em-mail the image of the chart, the chart title and a list of current selections. Make sure you have configured your device for e-mail.

- Bookmarks and Current Selections buttons are located on a common page. View the page by pressing the button to the upper right corner.
- When viewing a list box, shake the device to clear the selections shown in the list box.
- Move to horizontal or vertical to change perspective.
- Pan and zoom are supported.
- Text in text objects is displayed. Text objects are presented last in the cover flow.
- From the information screen you can access Help/Demo files and System details for support. There is an option to send an e-mail to support@qlikview.com. The message, which can be edited, contains information about the client software and the device.

Supported objects

The following objects are supported:

- List box
- Text object
- Charts
- Current selections
- Limited support* for:
- Pivot table**
- Straight Table***
- Table box

*Supported features are:

- Selection in table cells (single tap on a cell)
- Re-order table columns (tap on the header and drag the column)
- Sort a column (double tap on a column header)
- Links in tables
- Expression formatting
- Calculation conditions
- Table gridlines
- Table styles

Headers are locked while scrolling, there is limited column resizing.

**Supported features:

- Partial sums
- Expand and collapse by tapping the + symbol and the - symbol

***Supported features:

- All types of gauges as data representation for expressions
- Linked images as data representation for expressions
- Mini charts as data representation for expressions
- Subtotals
- Links

The totals row is always locked on top.

Design Guidelines

There are a few things to consider when developing applications that should be viewed on the iPhone client.

- Group Sheet Objects into Sheets so that they make sense without requiring a screen position. However, do not add too many Objects per Sheet. A maximum of 10 objects and 4 charts. (Charts take up the most resource.)
- All objects should be maximized.
- Charts should have white backgrounds.
- Charts should be brightly colored, and colors should match on a sheet.
- Charts should have axis titles.
- Charts should not have cycle groups or scroll bars.
- No need for Current Selections or Bookmarks objects on the sheets. However, you should assign some interesting bookmarks to be saved in the document.
- Make use of location services by adding two listboxes, preferably to a separate sheet, one name "longitude", and one name "latitude".
- Consider reducing data before distribution to reduce data traffic.
- Keep in mind that not all fonts are supported by the iPhone.

4 The QlikView BlackBerry and Java Mobile Clients

General

The BlackBerry and Java Mobile clients are both based on Java. The main difference between them is that while the Java Mobile client is restricted to only use the Java ME standard MIDP 2.0, the BlackBerry client uses BlackBerry native api for the User Interface.

The client connects to a QlikView Server, using either a socket connection (QVPJ) or HTTP/HTTPS, if tunneling is available on the server.

3G communication or WiFi is recommended. Connection speed is the most important factor for the performance of the mobile clients.

The client will, of course, be most effective on larger screens. If a touch screen is available it will be used, but almost all functionality is available even from a phone with just a numeric keypad.

Communication

The Java Mobile client communicates with the QlikView Server using the QVPX protocol. The protocol is xml based and used by both the Ajax Zero-Footprint client and the iPhone client. With the support for QVPX, a protocol designed for communication over HTTP, the performance is improved when running over HTTP or HTTPS compared to the older QVPJ protocol.

To use QVPX you must enable the setting **Use XML interface** in the Server setup screen.

The support for QVPJ will be removed in the future. QVPJ communicates either directly over a socket or via the tunneler using http or https. For authorization it uses an https connection to the QlikView Server to get a ticket.

Setup

To install the client use a browser for installation (by simply browsing to the JAD file). No special setup is needed on the QlikView Server, but if you want to use authorization you will need to configure https on your server with a certificate that is supported by your device. A QlikView Server Enterprise license is required to support the mobile client.

Configuration with JAD Files

JAD (Java Application Descriptor) files are part of the Java Mobile Edition framework. They contain some parameters that are mandatory for all Java ME applications and some QlikView specific parameters, e.g., what host to connect to, etc. They can be customized in a customer installation to point directly to the customer's host.

In the JAD file you can set the following parameters:

Server configuration parameters:

The first parameter should always be the DNS name or IP address, the second parameter, which is optional, should be the Server name. The user name and password cannot be pre-configured. The user will be prompted with a log-in dialog when connecting to the Server.

XML	Use QVPX xml interface
ANONYMOUS	Do not log in
SOCKET	Use socket for communication
HTTP	Use http communication
HTTPS	Use SSL communication
LOGINHTTP	Login over http
DIR=zzz	Set directory zz
READONLY	Do not allow the user to modify or remove Server configuration.

Up to ten QlikView Servers can be configured with the parameters QV-HOST and QV-HOST1 to 9, for example:

QV-Host: demo.qlikview.com;QlikView

demo;XML;ANONYMOUS;READONLY;DIR=mobile

QV-Host1: mobilebetaeu.qlikview.com;Beta EU;SOCKET;HTTP

General parameters:

DEBUG	Enable debug mode
DPI=xx	Set dpi to xx
SINGLE	Single object mode is default
STDMENU	Use standard menu (only for Java ME phones with touch functionality)
NOSTDMENU	Do not use standard menu (only for Java ME phones with touch functionality)
RECENT=y	Number of recent documents to save (0=no recent documents)
FORCE_MDS	Force MDS communication (only BlackBerry)
FORCE_TCP	Force direct Tcp communication (only BlackBerry)
FORCE_WAP2	Force WAP2 communication (only BlackBerry)
USE_WLAN	Use WLAN (only BlackBerry)
NOCOMPRESSION	Do not use compression

The communication path from the device to the server is a little more complicated than the standard client communication path, but basically it will go through the telephone operators network and over Internet to the

QVS host. If you use WiFi, which many modern devices support, and your QVS host is on the same local network as your wireless router you will, however not go over the internet.

Blackberry devices have a more complicated communication infrastructure, where communication is encrypted between the device and Blackberry BES server, if you use one of those. Blackberry also do not use JAR files but have their own JAD file format.

To set up Internet connection with your phone you might need to download settings from your operator. For BlackBerry phones you may need to enter settings by hand, depending on your service provider.

If you are installing from the download link and have an existing version already installed, you might need to clear your content cache. If you haven't done this you might get an error message 'JAR size mismatch'.

Centralized Configuration in BlackBerry

The BlackBerry client can be configured centrally. The configuration is done through the BlackBerry IT Policy. The administrator must log in to the BlackBerry Administration service and define IT Policy Rules. Add the setup parameters, QV-Setup, QV-Host or QV-Host1 to 9. The parameters the administrator can configure will be displayed under User Defined. The possible parameters are the same as those available for a JAD file (see above). All parameters configured in BlackBerry IT policy are automatically read only.

Changes in the IT Policy will automatically propagate to the clients.

Note! Settings made in the BlackBerry IT Policy will overwrite any parameters set in a JAD file!

Authorization

If authorization is enabled in the setup screen the mobile client will ask for username and password. It will then try to get a ticket from the QlikView Server. For this to work the server must be configured with SSL support.

Application Flow

The client shows QlikView documents in much the same way as the desktop clients, but on a screen that is considerably smaller. It is also limited by the font handling of Java ME, where only three, pre-defined font sizes (small, medium and large) are available. This means that text sizes might be different from the desktop clients and some objects might not work. Another limitation is on image sizes, where images that exceed the memory capacity of the device will not be displayed. They will, however be transferred from the server and can make opening the documents slow. When designing documents for the mobile client you should avoid using large images. Smaller images will work fine.

When you start the application you get a list of all your configured Servers. If you have opened documents before you also get an entry called **Recent documents**. When you click on a Server, the list of documents is displayed below the Server. In the settings form you can also see some information about your device and version of the QlikView Mobile client.

General Features

- The documents **Recent Documents** list can be removed
- **Server** name can be added in the Server settings
- A logo can be displayed on the documents

Supported objects

The following objects are supported:

- List box
- Table box
- Charts
- Text objects. Transparent text objects are supported in the BlackBerry client.
- Line/Arrow
- Button
- Current selections
- Statistics box
- Straight table
- Multi Box (read-only)
- Pivot table

Drawings of the objects are simplified compared to the desktop clients. Columns cannot be moved or resized.

Supported operations

The following operations are supported:

- Selection
- Clear
- Lock
- Unlock
- Back
- Forward
- Search
- Fast Change
- Apply bookmark
- Minimize
- Maximize/Restore
- Cycle and drill down
- Table sort
- Creating bookmarks
- Zoom (only newer BlackBerry and Symbian devices)
- Open URL. Depending on device the following URL types are supported:
 - http/https
 - tel
 - sms
 - mailto
- Pivot tables (pivoting is not supported. Expand/collapse is supported)
- Drill-up in list boxes and tables
- Cycle in list boxes and tables
- Tree view in list boxes
- URLs in tables
- Actions in line/arrow object

Note! Pivot tables are not supported in QVPJ!

Keyboard mapping phone keyboard

1: prev object	2:up	3:next object
4:left	5:select	6:right
7:prev tab	8:down	9:next tab
*:clear	0:back	#:forward

Arrow keys or scroll wheel scrolls the display, 2-4-6-8 moves within list boxes and tables. Other operations are available in the menu.

Keyboard mapping qwerty

Q:prev tab	W:	E: prev object	R:	T:up	Y:	U:	I: next object	O:	P:next tab
A:back	S:	D:left	F:left	G:	H:right	J: right	K:	L:	Backspace:clear
	Z:forward	X:	C:clear	V:down	B:down	N:	M:start select		Enter:Select

The Escape key on BlackBerry is mapped to the Back function in QlikView. If no more steps backward are available, the document is closed. Trackball scrolls the display, T-F-H-V moves within list boxes and tables. Many functions are mapped to several keys, to support Blackberry with small keyboard, like Pearl.

Touch screen support

Touch screen is supported:

- Selection is made by pressing the screen
- Setting the pointer in the background, on a non-focusable object or in the caption and dragging scrolls the screen
- Clicking on maximize or restore icon in object caption will maximize or restore object
- To highlight an object you must double-click the object.

On phones without touch screen a cursor is used and clicking middle soft key or trackball will simulate pointer press or release, depending on context.

Design Guidelines

-
- Keep objects small. The screen is scrollable, but objects that are larger than the screen are generally not a good idea.
- Verify that the font sizes you use work on your device. Small fonts might be displayed larger on the device, which might mean that texts will be cut off. The exception to this rule is graphs, which are rendered on the server, so scalable fonts can be used. Keep texts short if possible.
- Avoid buttons, if your users don't have a pointing device.

- Only use small images. A large image will take time to transfer to the device and make your document slow and you do not want to use your limited memory for large images. If the image is too big to fit into the devices memory it will not be displayed (but still take time to transfer).

Web Server for Mobile Downloads for QlikView

Note! The set up below is only one way of setting up a site for downloading the mobile client!

Prerequisites

In order to make the mobile client downloadable from a phone, the following prerequisites should be fulfilled:

- The Midlet files (Java files)
- An URL used as the download location
- Optionally a redirect site (Sample source code is included)

If Microsoft IIS is used as web server, the file getticket.asp must be placed in the Script folder. It is installed together with the feature **Microsoft IIS Support**.

Configuring the Web Server

Configure the following mime-types:

*.jad text/vnd.sun.j2me.app-descriptor

*.jar application/java-archive

*.cod application/vnd.rim.cod

This step must not be skipped!

Modifying the Midlet Configuration Files

Extract the Midlet on the local harddrive. The structure will look as follows:

QvJME.jar

QvJME.jad

bb\QvQlikView.jad

bb\QlikView-1.cod

bb\QlikView.cod

Upload to the WebServer

Upload the Midlet files to webserver.

Configuring the redirect site (optional)

Depending on the types of phones accessing the download site, they might be redirected to different sites depending on the type. The two most common redirects are:

Non-mobile-phone => "Please use a phone to access this page"

Blackberry => bb\QvJME.jad

All other => QvJME.jad

Sample source code

```
if (Request.ServerVariables["HTTP_USER_AGENT"].Contains("Windows NT"))  
Response.Redirect("windowsuser.aspx"); else if (Request.ServerVariables["HTTP_USER_  
AGENT"].Contains("BlackBerry")) Response.Redirect("bb/qvjme.jad"); else  
Response.Redirect("qvjme.jad");
```

Test

Open the URL pointing at the redirect site (or directly on the QvJME.jad) in the mobile phone.

5 Developing Documents for Use with QlikView Server

General

The QlikView Server will read all qvw files created by QlikView 6.0 and later. Files with older file formats must be converted to a newer file format before used with the QlikView Server.

General limitations when working from clients

Regardless which client you use there are some things you cannot do when working with a document opened through QlikView server:

- You cannot create new documents
- You cannot access or execute the script
- You cannot access or change the macro module
- You cannot save changes to the document, beyond shared objects

Performance Considerations

Response time and memory usage both on the server and on the client can depend heavily on how your application is designed. Below are a few tips to bear in mind when building your application:

1. Each open chart and table will require extra memory in the server for each user connected. Minimized charts/tables as well as charts/tables not on the active sheet do not require extra memory. Each open chart/table must also be recalculated on the server after each change in the logical state, which could prolong response times. Therefore avoid opening charts and tables when it is not necessary, or try to spread them over several sheets.
2. Large bitmap pictures in sheet backgrounds or text objects may cause delays, as they have to be transferred from the server to the client. The bitmaps are compressed during transfer and are only transferred once per session, but nevertheless can cause substantial amount of data to be moved. Consider using colored sheet background and text objects with opaque background for better performance. If you have to use bitmap sheet backgrounds, it is more efficient to apply one background on application level than assigning the same background to each sheet.
3. Very small (e.g. 1x1 pixel) tiled background bitmaps can greatly reduce performance on the client, as they have to be rendered repeatedly by the Java Virtual Machine. Rather use medium size (e.g. 20x20 pixel) tiles or colored sheet background.

Document Configuration

Certain settings in the QlikView document are utilized for additional purposes when that document is opened under QlikView Server. The following considerations are suggestions for making the QlikView Server experience better for your clients.

1. The Object caption Title Text is used in any document list of objects in the collaboration dialogs. If the caption Title Text is not set, the object ID will be used. Be sure to set the object caption Title Text on all objects, including buttons, line arrow, and text objects, which do not use a default title. You can set the object caption Title Text on the object Properties, Caption dialog.

2. Expression Labels are used in the collaboration dialogs. The default Expression Label is the expression itself, which may not be descriptive. Be sure to set all Expression Labels in chart objects to a descriptive name.
3. Avoid using large images for the mobile clients, since the transfer of these images can cause response time delays for these clients. Images that exceed the memory capacity of the device will not be displayed. They will, however be transferred from the server and cause delays in the opening of the document. Smaller images should perform fine.

Security and Access restriction

It is important that information is distributed only to those with access rights to it. Since QlikView makes the previously cumbersome process of retrieving information a very simple task, it is obvious that an access restriction mechanism is necessary. This can be done when creating the application. For more information on this, see the QlikView reference manual.

When a QlikView document is published on a QlikView Server, it is just as important to protect confidential data and prevent unauthorized access as it is when running the standard QlikView. QlikView Server therefore supports all security features built into QlikView. In addition, it is possible to use the Windows NT file security or DMS authorization to control access to specific documents. This can be done also without using any of the QlikView built-in security features.

QlikView File Security

QlikView and QlikView server can be instructed to test for a number of user properties before letting the user open a document. (For more information on this, see the QlikView reference manual.) Possible fields are:

USERID: A field that should contain a valid user ID. The client will open a dialog asking for user ID.

PASSWORD: A field that should contain a valid password. The client will open a dialog asking for Password.

SERIAL: A field that should contain a number corresponding to the QlikView serial number. The server will check the Server Serial Number and compare that to the valid ones listed in the QlikView document. In other words, the Serial Number check is a very blunt security instrument for documents that are to be published: Any client will be allowed to open a document on the valid server.

NTNAME: A field that should contain a string corresponding to a Windows NT Domain user name or group name. The server will ask the operating system if the user named in the QlikView document corresponds to the user logged on, or to a group to which the logged on user belongs.

NTDOMAINSID: A field that should contain a string corresponding to a Windows NT Domain SID. The server will ask the operating system if the Windows NT Domain named in the QlikView document corresponds to the Domain to which the logged on user belongs.

NTSID: A field that should contain a Windows NT SID. The server will ask the operating system if the NT SID named in the QlikView document corresponds to the NT SID of the user logged on, or to a group to which the logged on user belongs.

Only if all the required tests are passed, will the user be able to open the document.

The Windows File security described above and the three last tests are of course very similar. The difference is that for the former, the security information is fetched from the file system, whereas for the latter, it is fetched from within the QlikView file. Both are, however, compared to the logon information from the operating system.

Dynamic Data reduction

QlikView Web Server supports a feature that can hide some part of the data (field values) in a document from the user based on the login information. Only the information that the user has the right to see will be shown.

The selection of values to be shown is controlled by means of having one or more fields with common names in the data and in the access restriction data. Only the field values logically connected to the user logged on will be shown.

This is a very efficient way to hide information to which the user does not have access.

For more information on how to build such an application, see the QlikView reference manual.